

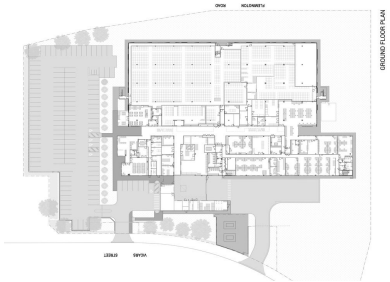
Global Industrial Architecture

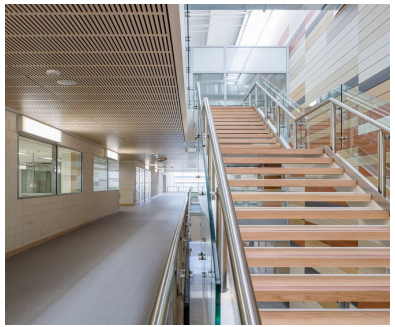
Maisie Richter

Selected Case Study: National Archives Preservation Facility (Australia)



The National Archives Preservation Facility (NAPF) serves as a repository for more than 15 million paper and audiovisual records, focusing on their storage, management, and preservation. Guided by the principle of "Your story, our history," the facility's architectural design embodies its role as the guardian of the nation's shared memory. It symbolizes the essential functions of archiving and provides a dedicated space for archivists. Central to its design concept is the depiction of the passage of time, reflecting both the past and the future.





Architects: May + Russell Architects

Area: 18000 m²

Built Year: 2017

Site Analysis

Typology - Ecological Function: Ecological function of vegetation on the site, such as providing habitat for wildlife, improving soil stability, enhancing biodiversity, and mitigating environmental impacts such as erosion and stormwater runoff for the facility.

Address: 30 Vicars St, Mitchell ACT 2911

Pedestrian circulation through car park.

Pedestrian /worker circulation around facility.

Sun Path and Climate:The ACT typically sees daily high temperatures in summer ranging from 26–28°C in the northern regions to approximately 18–20°C in the alpine areas. Winter brings average low temperatures ranging from 2–4°C in the north to around -4 to -2°C in the southern parts and the alpine regions. Throughout the year, this territory vast and distinct seasonal changes.

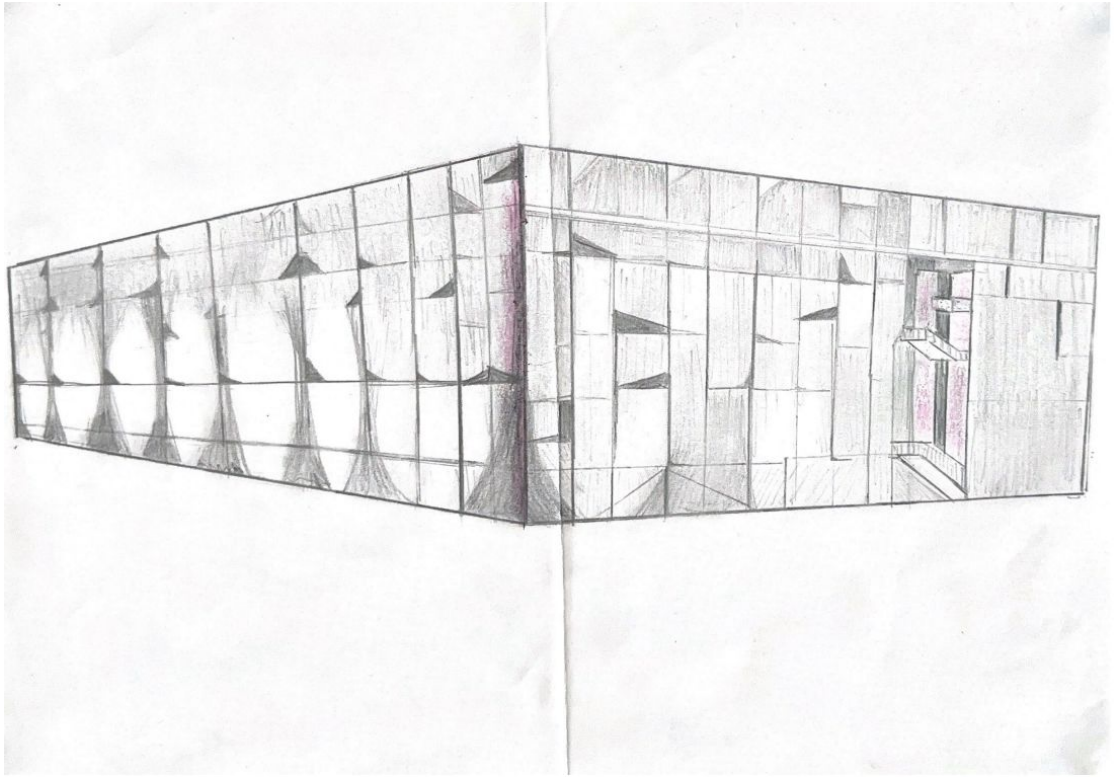
Analysis of Parking Facilities: The parking options are available next to the facility, including on-site parking areas and nearby public parking lots and garages. There is a large capacity of space, the layout enables more e=than enough access into the building.

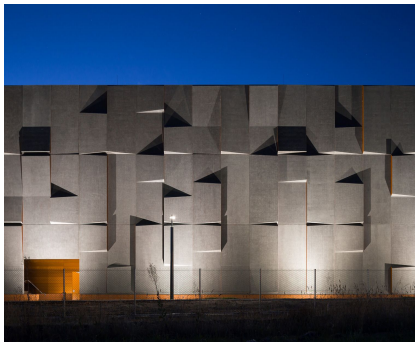
Road Access: The facility is accessible by road including freeways and local roads, with designated entrances and parking areas for visitors and staff. Roads leading to the facility may are well-marked and easily navigable, allowing for convenient access by car or other private vehicles.

Natural Light: The site benefits from abundant natural light throughout the day due to its open exposure and minimal obstruction from surrounding buildings or vegetation. During daylight hours, sunlight penetrates deeply into the site, creating well-lit outdoor spaces and reducing the need for artificial lighting.Although, as it is not obstructed by vegetation during the hottest part of the day, it may require other elements to the building to ensure that it does not get too hot within the building.



Perspective Drawing







Plans and Timeline for my model of the National Archives Preservation Facility

Shopping List:

- Foam Board (A1 5mm) $9.54
- CardBoard (3mm 900x600mm) $4.96
- Clear Contact Paper (5mm) $5.85
- Clear Gorilla Glue (51ml) $12.95
- Gorilla Emergency Repair Tape, 8.2 Meter Length, Clear $19.43
- Coloured Markers $2.00
- Quill A4 280gsm Kraft Board 25 Pack $31.29
- Quill A3 Flute Board Assorted 5 Pack $13.49
- Fiskars Softgrip Craft Knife $17.39

Utensils at home:

- Scissors
- Ruler
- Stencil
- Stanley Knife
- Sharpener
- Eraser
- Pencils

Plan:

1. Utilizing printed drawings to trace and replicate shapes and structures onto foam board.
2. Employing kraft board as the foundation/base material for the model.
3. Crafting the model by shaping foam board into desired forms.
4. Colour in the cut outs to replicated colour pallet.
5. Adhering primary structural components together using glue.
6. Incorporating clear contact paper sheeting for windows.
7. Utilizing silver flute board to represent metal roofing materials
8. Allocate additional days for the adhesive to fully dry.
9. Completion of Model.

Time Management	Date	Completion Status
Materials	31/03/2024	✅
Started Marking out measurements on Foam Board	31/03/2024	✅
Cutting out parts of model	31/03/2024	✅
Assembling Model	02/04/2024	
Final touches and colour selections	04/04/2024	

Reflection and plan for my Model

As I delve into constructing my model, I've come to realize that achieving precision and authenticity demands thorough groundwork and unwavering commitment to each phase of the endeavor. The journey commences with extensive research, delving into architectural blueprints and reference materials to grasp the intricate dimensions and characteristics of the structure, ensuring meticulous adherence to scale. This foundational understanding informs my material selection, ensuring alignment with the intended scale and level of intricacy. Armed with precise tools like rulers and cutting implements, I meticulously measure and shape each component, striving for immaculate lines and exact angles. The assembly process unfolds methodically, commencing with the groundwork and steadily layering components while regularly consulting my research to uphold fidelity. Patience becomes paramount, allowing adequate drying time for adhesives and reinforcing structural integrity as needed. Throughout this journey, I continuously reference my research, making adjustments to maintain faithful representation. With each stroke and detail, I endeavor to encapsulate the essence of the structure, transforming it into a tangible manifestation of my creative vision. Through this hands-on experience and unwavering dedication to authenticity, my model transcends mere craftsmanship—it becomes a testament to my passion for architectural precision and meticulous detail, all carefully rendered to scale.

Short Reflection/ Graphic Style Plan

Choosing Global Industrial Architecture was the focus of our project, a decision fueled by a blend of personal interests and academic aspirations within our group (Maisie, Ornela, Sam and Noor).

In our collaborative efforts, we consciously opted for foam board as the primary material for our model, driven by a collective appreciation for its minimalist aesthetics and its ability to deliver a refined finish. Recognizing the importance of clarity and simplicity in architectural representation, we found that foam board offers the ideal canvas to accentuate the core elements of our design without unnecessary embellishments. Its adaptable nature allows us to effortlessly sculpt and mold it to mirror the intricate details and contours of our vision, ensuring an accurate and visually striking portrayal. By selecting foam board, we ensure that our model will exude a sense of coherence and sophistication, effectively capturing the essence of our design with precision and finesse.

Industrial architecture, with its fusion of functionality, innovation, and historical relevance, offers a captivating narrative of human creativity and progress. It transcends mere structures, delving into the narratives of societal needs, technological advancements, and the dynamic evolution of landscapes. This exploration not only enriches our understanding of architectural evolution but also serves as a mirror reflecting global trends and challenges in construction and engineering. Each industrial site, influenced by its unique cultural, environmental, and economic context, offers a rich tapestry of architectural styles, construction methods, and materials to study and appreciate.

Moreover, our interest in industrial architecture extends beyond its aesthetic and historical value. It presents an intriguing canvas for exploring sustainability and adaptive reuse a subject increasingly pertinent in contemporary construction discourse. By examining successful adaptive reuse projects, we aim to glean insights into repurposing existing structures, minimizing waste, and fostering resilient, environmentally conscious built environments.

Individually, our fascination with industrial architecture varies, reflecting diverse interests within the realm of construction. Some of us are drawn to the historical narratives embedded in industrial sites, while others are captivated by the potential of sustainable design principles or the integration of technology in modern industrial facilities. This project provides an avenue for each of us to pursue our unique passions while contributing to a collective endeavor.

Selecting the National Archives Preservation Facility as my individual case study within our group's broader exploration of industrial architecture around the world was a decision rooted in a blend of personal fascination and strategic alignment with our overarching project goals.

On a personal level, the National Archives Preservation Facility captivates me due to its unique intersection of history, architecture, and function. It's not just a repository of documents; it's a custodian of our collective memory, safeguarding the records that tell the story of our nation's past. The facility's architectural design, with its emphasis on preservation and symbolism, speaks volumes about the importance of heritage conservation in a rapidly changing world.

Moreover, choosing the National Archives Preservation Facility aligns seamlessly with our group's larger mission of exploring industrial architecture on a global scale. While it may seem distinct from traditional industrial structures like factories or warehouses, the facility embodies the principles of industrial architecture in its design, construction, and purpose. It showcases how architectural innovation and engineering prowess can converge to create a functional yet aesthetically compelling industrial space dedicated to the preservation of cultural heritage.